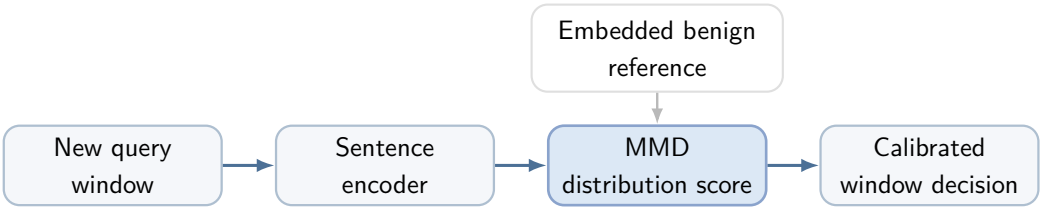
Proposed project: continuous LLM API monitoring



Reusable artifacts

- Query data and baseline code.
- Fixed BGE-small embeddings.
- Benign-only calibration.

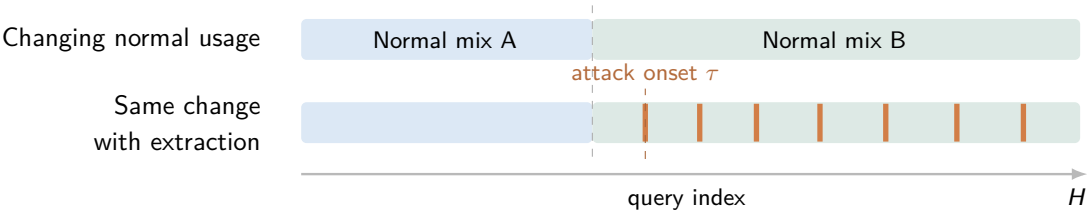
Scope of the existing study

- Offline traffic windows.
- Mixed and adaptive attacks.
- Reference contamination.

Liu, Guo & Dong, 2026. Official code and data instructions. Published setup: query-only, offline window decisions.

The research question is sustained monitoring

How early can we detect extraction under changing normal usage, at the same false-alarm budget?



Repeated decisions

Window-level errors can accumulate over a monitoring run.

A changing reference

Adaptation may reduce drift alerts and also absorb extraction traffic.

Motivation: Liu, Guo & Dong, 2026, Limitations. The query-order schematic shows a proposed replay design, not measured outputs.

Controlled replay under benign traffic changes

	No extraction traffic	Sparse extraction traffic
Stable normal usage	False alarms over a full run.	Detection within a query budget.
Changing normal usage	Alerts caused by benign change.	Detection while normal usage changes.

Initial pairs: `model_leeching` and `Meaeq_SST-2`

Split source queries before constructing streams. Reuse each replay across methods. Measure delay in query counts.

Official code and data instructions. The README illustrates id, index, and query. Production timestamps and account histories are not established by this artifact.

Alarm rules and sequential baselines

Comparison	What it tells us
Published MMD setup	Checks a restricted reproduction.
MMD + run-level threshold	Isolates calibration for repeated decisions.
MMD + accumulation	Tests evidence combined across windows.
KCUSUM	An established kernel sequential detector.
CALM	An established sequential-calibration approach.

Shared inputs: same query pools, replays, decision times, and calibration budget.

Keep the original threshold as a reproduction reference; calibrate eligible methods to the same run-level target for the main comparison.

Liu, Guo & Dong, 2026; Flynn & Yoo, KCUSUM, CDC 2019; Cobb et al., CALM, AISTATS 2022.
KCUSUM and CALM use their native update and calibration schemes.

Timely detection at a matched false-alarm target

Benign run

$$\Pr_0(T \leq H)$$

At least one false alarm in H queries.

Attack run

$$\Pr(\tau \leq T \leq \min(\tau + B, H))$$

Detection within B queries after onset τ.

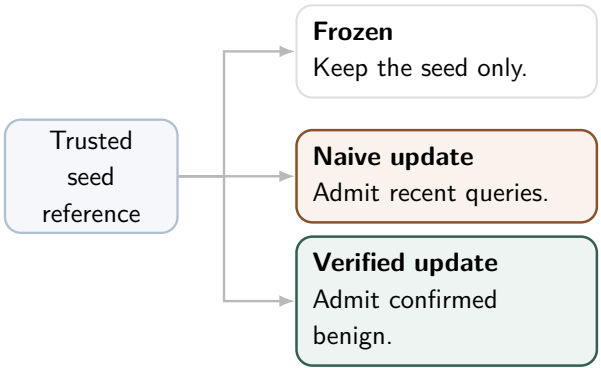
Primary comparison: calibrate on benign replays including the declared drift family; evaluate on held-out runs. Also report delay and late or missed attacks.

First-alarm protocol

T is the first alarm index; $T = \infty$ means no alarm. An alarm before onset is a false alarm and fails the timely-detection criterion.

Proposed metrics. H: monitoring horizon. Stable-only calibration followed by unseen drift is a separate stress test, without a matched-budget claim.

Reference updates under explicit information assumptions



Proposed hypothesis

Updates may reduce alerts from benign change, while unverified updates may hide persistent extraction.

Information requirement

Trusted updates need verified benign queries. Hidden test labels define an oracle diagnostic only.

Reference contamination is evaluated in Liu, Guo & Dong, 2026. Proposed comparison: full-run update policies under drift, with a shared verification budget.

Evidence required for a research contribution

Evidence from the study	Next step
A failure survives matched calibration and existing sequential baselines.	Investigate a targeted monitoring method.
Reference updates introduce a further failure under explicit assumptions.	Study the update rule and information needs.
Simple calibration or existing methods resolve the difficulty.	Report it; narrow or revise the question.

Check whether the pattern survives a second data pair and a new source split.

Decision criteria for a proposed study. No new performance results or novelty claims are assumed here.

Study scope and expected outputs

Query-only monitoring under declared benign changes.

Experimental scope

- Controlled query-only replay.
- Fixed query representations.
- Declared drift and verification.

Expected outputs

- Reproducible replay configurations.
- False alarms and timely detection.
- Evidence for a focused method.

Residual failures determine whether a new monitoring mechanism is needed.

Proposed study. No new detector performance or deployment guarantee is assumed.

Replay protocol and information boundaries

Data separation

- Group duplicates and common source items before splitting.
- Separate reference, development, calibration, and test pools.
- Fit topic groups on development data only.
- Freeze horizon, window, stride, and replay design before testing.

Interpretation

- Reordering a fixed corpus adds no independent queries.
- Calibrate the full alarm and update policy.
- Verified updates require an explicit information source.
- Query-only monitoring needs assumptions about benign change.

Identical observable query distributions cannot reveal different intentions from query content alone.

Proposed protocol. Calibrate the complete alarm policy before held-out evaluation.

Related work and the proposed increment

Existing work	What it contributes	Role in the proposed study
Liu, Guo & Dong 2026	MMD on query windows with benign calibration.	Released starting point.
Flynn & Yoo CDC 2019	KCUSUM: kernel sequential change detection.	Accumulation baseline.
Cobb et al. AISTATS 2022	CALM: sequential false-detection calibration.	Calibration baseline.

Proposed increment: continuous API replay, followed by any unresolved reference-update failure under explicit information assumptions.

MMD: arXiv 2606.05725 KCUSUM: arXiv 1903.01661 CALM: PMLR 151, 226–239.

Provenance and applicability of reference updates

LangSkills: traceable artifacts

- Records source references and generation metadata.
- Validates skill structure and metadata.
- These checks do not establish scientific validity.

HazardWeaver: explicit conditions

- Links applicability conditions to evidence.
- Distinguishes satisfied, violated, and unresolved conditions.
- Unresolved required evidence blocks immediate use.

Application to this proposal: record reference provenance, verification evidence, and the benign-change regime supported by calibration.

This documentation principle is proposed here. Neither source establishes its effectiveness for API extraction monitoring.

LangSkills (source artifacts and validator). Zhu et al., HazardWeaver, 2026 (Sec. 3.1 and Appendix J).